Seminar topics

Advanced Computer Graphics, 2025/26

1 Introduction

The document describes the seminar topics for the Advanced Computer Graphics course given in 2025/26. The seminar work must be completed during the spring semester and defended prior to the final deadline. Students must defend the seminar at the laboratory exercises and present it at the final presentation session in front of the class. Upon successful completion, the seminar may contribute up to 30% to the final grade. For a positive grade, the seminar must be graded with at least half the points.

The expected outputs of each seminar are the seminar report and code repository published on GitHub under the CC-BY, BSD or MIT license. The report must be submitted through the FRI Učilnica. It must be written as a research paper in accordance with the **Eurographics LaTeX template**. The report must present the problem, the solution, and the results.

2 Seminar Topics

2.1 Light propagation volumes for global illumination

Accurate and efficient computation of global illumination is a significant challenge in rendering. The appearance of materials is heavily influenced by the surrounding environment, making the quality of rendering depend on effective environment sampling. While path tracing can produce accurate results, it is often too slow for real-time execution. A common practical solution is to precompute radiance at fixed points in space and use these values during rendering to avoid costly computation for every light ray. The points are often placed on a grid to facilitate sampling, and the resulting structure is called a light propagation volume. In this seminar, you will explore light propagation volumes and implement the approach in a prototype application. You will evaluate the results both qualitatively and quantitatively and compare them to path tracing.

References:

[1] Kaplanyan, Anton, and Carsten Dachsbacher. "Cascaded light propagation volumes for real-time indirect illumination." Proceedings of the 2010 ACM SIGGRAPH symposium on Interactive 3D Graphics and Games. 2010.

[2] Kaplanyan, Anton. "Light propagation volumes in cryengine 3." ACM SIGGRAPH Courses 7.2 (2009).

[3] Billeter, Markus, Erik Sintorn, and Ulf Assarsson. "Real-time multiple scattering using light propagation volumes." I3D 12 (2012): 119-126.

2.2 Procedural generation of tree log models

In this seminar, you will develop a program for procedurally generating 3D models of tree logs based on user-defined parameters. The program should take inputs such as log length, diameter, eccentricity, surface roughness, curvature, tapering, number and shape of gnarls, and other relevant characteristics. Using these parameters, the program should generate a detailed 3D model of a tree log, including a realistic texture that reflects the natural variations found in real wood. The goal is to create a flexible system capable of producing diverse and lifelike log models suitable for use in rendering, simulations, or game environments.

References:

[1] Baša, Mirko. Napake lesa: priročnik za prepoznavanje napak okroglega lesa: opisi najpomembnejših napak lesa, njihovo merjenje in določanje kvarnega vpliva na kakovost. Gozdarski inštitut Slovenije, založba Silva Slovenica, 2020. https://wcm.gozdis.si/downloads/Napake_lesa_Net4Forest_web.pdf

[2] Leban, Irena. "NAPAKE V LESU." https://cpi.si/wp-content/uploads/2020/08/5-NAPAKE_LESU.pdf

2.3 Gaussian splatting rendering denoising

In this seminar, you will explore how to denoise Gaussian splatting renders. You will start with an order-independent Gaussian splatting implementation that uses stochastic transparency sampling to eliminate the need for sorting but introduces visible noise into the final image. The renderer produces both an albedo image and a depth map, which can be directly used as inputs for a denoising pipeline. Your task will be to pass the renderer's output through a real-time denoiser to remove the noise and improve image quality. You can use an existing denoising method or design and train your own approach.

References:

[1] Mara, Michael, et al. "An efficient denoising algorithm for global illumination." High Performance Graphics 10.3105762-3105774 (2017): 2.

[2] Chaitanya, Chakravarty R. Alla, et al. "Interactive reconstruction of Monte Carlo image sequences using a recurrent denoising autoencoder." ACM Transactions on Graphics (TOG) 36.4 (2017): 1-12.

2.4 Gaussian splatting model compression

In this seminar, you will explore techniques for compressing Gaussian splatting models to make them more memory efficient. You will begin with a baseline Gaussian splatting representation that achieves high visual quality but consumes a lot of memory due to the associated attributes. Your task will be to investigate and implement Gaussian model compression strategies such as quantization, attribute sharing, delta compression, or more advanced learning-based compression methods, while preserving rendering quality as much as possible. You will evaluate the trade-offs between compression rate, visual fidelity, and rendering performance, and assess how different approaches impact real-time rendering capabilities.

References:

[1] Bagdasarian, Milena T., et al. "3dgs. zip: A survey on 3d gaussian splatting compression methods." Computer Graphics Forum. Vol. 44. No. 2. 2025.
[2] Ali, Muhammad Salman, et al. "Compression in 3d gaussian splatting: A survey of methods, trends, and future directions." arXiv preprint arXiv:2502.19457 (2025).
[3] Cao, Chao, Marius Preda, and Titus Zaharia. "3D point cloud compression: A survey." Proceedings of the 24th International Conference on 3D Web Technology. 2019.

2.5 Gaussian splatting model simplification

In this seminar, you will explore methods for simplifying Gaussian splatting models by constraining the number of splats to achieve a desired level of detail. You will start with a high-quality baseline model containing a large number of Gaussians. Your task will be to develop and evaluate strategies for reducing the number of splats while maintaining a specified visual quality target. This may involve techniques such as importance-based selection, clustering, and merging of Gaussians. You will analyze the trade-offs between model size, rendering performance, memory usage, and visual fidelity, and assess how different simplification strategies affect real-time rendering and scalability. You will use that knowledge to design a level-of-detail scheme.

References:

[1] Fang, Guangchi, and Bing Wang. "Mini-splatting: Representing scenes with a constrained number of gaussians." European conference on computer vision. Cham: Springer Nature Switzerland, 2024.
[2] Zhang, Yangming, et al. "Gaussianspa: An" optimizing-sparsifying" simplification framework for compact and high-quality 3d gaussian splatting." Proceedings of the Computer Vision and Pattern Recognition Conference. 2025.

2.6 Gaussian splatting in VR

In this seminar, you will explore the use of Gaussian splatting for virtual reality (VR) applications. You will begin with a baseline Gaussian splatting renderer and extend it to support stereoscopic rendering in a VR environment. Your task will be to adapt the rendering pipeline to meet the specific requirements of VR, including real-time performance, low latency, high frame rates, and consistent visual quality across both eyes. This may involve optimizing data structures and shaders, handling stereo camera setups, integrating with a VR runtime, and addressing challenges such as view-dependent effects and motion stability. You will evaluate the resulting implementation both quantitatively and quantitatively.

References:

[1] Jiang, Ying, et al. "Vr-gs: A physical dynamics-aware interactive gaussian splatting system in virtual reality." ACM SIGGRAPH 2024 conference papers. 2024.
[2] Tu, Xuechang, et al. "Vrsplat: Fast and robust gaussian splatting for virtual reality." Proceedings of the ACM on Computer Graphics and Interactive Techniques 8.1 (2025): 1-22.

2.7 Seamless transition between Gaussian splatting and mesh rendering

In this seminar, you will explore techniques for enabling seamless transitions between Gaussian splatting and mesh-based rendering. You will use a high-quality Gaussian splatting representation for detailed close-up views, and a simplified mesh representation for low-detail, distant views. Your task will be to implement a technique for smooth transitions between the two representations as the viewing distance or required level of detail changes. This may involve screen-space blending or alignment of geometric and appearance information to avoid visual artifacts such as popping or discontinuities. You will discuss how such a hybrid approach can improve scalability while maintaining high image quality across different levels of detail.

References:

[1] Guédon, Antoine, and Vincent Lepetit. "Sugar: Surface-aligned gaussian splatting for efficient 3d mesh reconstruction and high-quality mesh rendering." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2024.
[2] Shao, Zhijing, et al. "Splattingavatar: Realistic real-time human avatars with mesh-embedded gaussian splatting." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024.

2.8 Void space visualization

In this seminar, you will explore techniques for void space visualization in point cloud rendering. Each point in a point cloud carries a set of attributes such as color, size, motion, and depth (available after rendering). However, rendering each point as a single pixel is often insufficient, as it leaves gaps between samples and does not produce a visually coherent image. You will investigate methods for interpolating point attributes in screen space to produce a continuous image. The focus will be on efficient interpolation techniques that enable real-time performance even in high resolution, such as multigrid. Your task will be to implement a render and interpolation pipeline that first renders the point cloud naively and then propagates the attributes into empty regions. You will assess how real-time void space visualization can improve the usability and appearance of point cloud renderings.

References:

[1] Kreiser, Julian, Pedro Hermosilla, and Timo Ropinski. "Void space surfaces to convey depth in vessel visualizations." IEEE Transactions on Visualization and Computer Graphics 27.10 (2020): 3913-3925.
[2] Erzar, Blaž, Žiga Lesar, and Matija Marolt. "Fast incremental image reconstruction with CNN-enhanced poisson interpolation." Václav Skala-UNION Agency (2023).
[3] Özbay, Ali Girayhan, et al. "Poisson CNN: Convolutional neural networks for the solution of the Poisson equation on a Cartesian mesh." Data-Centric Engineering 2 (2021): e6.

2.9 Image-based lighting

In this seminar, you will develop a real-time rendering pipeline that uses image-based lighting (IBL) to illuminate 3D models with realistic environmental lighting. You will implement a pipeline that processes an environment map for a given physically based material and uses it for real-time rendering. The pipeline should include preprocessing steps such as generating diffuse irradiance maps through convolution, creating specular reflection maps with varying roughness levels using importance sampling, and computing the split-sum approximation for the BRDF integration. During rendering, you will use these precomputed maps to efficiently evaluate both diffuse and specular contributions. You will evaluate the visual quality and performance of your implementation and analyze the trade-offs between accuracy and real-time performance.

References:

[1] Debevec, Paul. "Image-based lighting." ACM SIGGRAPH 2006 Courses. 2006. 4-es.
[2] Debevec, Paul. "Image-based modeling and lighting." ACM SIGGRAPH Computer Graphics 33.4 (1999): 46-50.

2.10 Neural texture compression

In this seminar, you will explore neural texture compression for reducing texture storage requirements in real-time rendering applications. Modern games rely heavily on high-resolution textures to achieve realistic material appearance, but even with standard block compression formats like BCn, the texture storage requirements can reach several gigabytes. Your task will be to implement a neural texture block compression system that uses small neural networks to encode texture data at lower bit rates than general-purpose compression formats. The system should process multiple textures belonging to a single material (such as diffuse color, normal maps, and other BRDF parameters) simultaneously, and store only the neural network weights. You will compare compression ratios, visual quality, encoding time, and sampling performance against BC1 and BC4.

References:

[1] Vaidyanathan, Karthik, et al. "Random-access neural compression of material textures." arXiv preprint arXiv:2305.17105 (2023).
[2] Fujieda, Shin, and Takahiro Harada. "Neural Texture Block Compression." MAM/MANER@ ESGR. 2024.

2.11 Smoothed-particle hydrodynamics for fluid simulation

In this seminar, you will implement a fluid simulation with smoothed particle hydrodynamics (SPH). SPH represents fluids as a collection of particles, where each particle carries physical properties such as position, velocity, density, and pressure. Your task will be to develop a basic SPH implementation including density estimation, pressure, viscosity, and external forces such as gravity. You will then render the simulation with surface reconstruction (e.g., marching cubes). You will evaluate the performance and visual quality of the simulation with respect to various input parameters (initial state, number of particles, etc.).

References:

[1] Ihmsen, Markus, et al. "SPH fluids in computer graphics." (2014).
[2] Harada, Takahiro, Seiichi Koshizuka, and Yoichiro Kawaguchi. "Smoothed particle hydrodynamics on GPUs." Computer Graphics International. Vol. 40. SBC Petropolis, 2007.
[3] Monaghan, Joe J. "Smoothed particle hydrodynamics." In: Annual review of astronomy and astrophysics. Vol. 30 (A93-25826 09-90), p. 543-574. 30 (1992): 543-574.

2.12 Rendering large graphs on the GPU

In this seminar, you will implement a high-performance 2D graph rendering framework using WebGPU, enabling real-time visualization of massive node-edge datasets in the browser. The framework should support interactive features such as node and edge selection, highlighting, filtering, and dynamic updates, while allowing extensive visual customization. The second goal is to provide an experience comparable to traditional SVG-based rendering systems, supporting graphical attributes including colors, borders, textures or images, transparency, and depth ordering. The resulting framework should be compared to other popular GPU-based graph rendering frameworks, such as D3 [1], Sigma.js, reagraph, and G6. This task is primarily focused on rendering, so the underlying data representation can be based on libraries such as D3 [1] or Graphology [2]. This task is strongly engineering-oriented and assumes prior knowledge of modern web technologies, computer graphics, WebGPU, and JavaScript/TypeScript development.

References:

[1] Bostock, Michael, Vadim Ogievetsky, and Jeffrey Heer. "D³ data-driven documents." *IEEE transactions on visualization and computer graphics* 17.12 (2011): 2301-2309.
[2] Guillaume Plique. (2021). Graphology, a robust and multipurpose Graph object for JavaScript. Zenodo. https://doi.org/10.5281/zenodo.5681257

2.13 Tree generation parameter estimation from point clouds

This seminar examines the usefulness of 3D tree mesh generators for reconstructing trees from sparse point clouds. It is suggested that you choose an existing tree generation tools (e.g. Blender's Sapling Tree Gen, Easy Tree, Modular Tree, or others). You will begin with a point cloud scan of a tree and a ground truth mesh or skeleton, then develop an algorithm or use neural network configurations that optimize tree generation parameters, mainly geometric parameters such as trunk width, branch distribution, etc., so that the generated mesh matches the point cloud as closely as possible. The resulting mesh will likely not fit the scanned tree completely, but it may provide a suitable starting point for more granular tree fitting in later stages. The results could potentially be applied to environment scans compression, modelling entire forests from LiDAR scans, supporting landscape reconstruction for visualizations and interactive 3D environments, biomass estimation, and various forestry applications. The final parameter estimation method would be implemented as an extension in Blender 3D modelling software.

References:

[1] Xia, Shaobo, et al. "Geometric primitives in LiDAR point clouds: A review." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 13 (2020): 685-707.
[2] Wang, Xuan, et al. "Oblique photogrammetry supporting procedural tree modeling in urban areas." *ISPRS Journal of Photogrammetry and Remote Sensing* 200 (2023): 120-137.

2.14 Tree mesh reconstruction from point clouds

The aim of this seminar is to examine one or two mesh reconstruction methods for point cloud representations of trees. First, a ground truth dataset should be obtained or generated, providing point clouds with corresponding mesh models. Each reconstruction method should be thoroughly analyzed and tested. The results should be compared with those of other approaches. If only one method is used, its results should be carefully examined to identify cases where the approach fails and to explore ways to address these issues and improve reconstruction. The resulting outputs would support the construction of detailed 3D models from environmental LiDAR or photogrammetry scans, with applications in games, animation, forestry (e.g. biomass estimation), environmental analysis, and related fields.

References:

[1] R. Wang, J. Peethambaran and D. Chen, "LiDAR Point Clouds to 3-D Urban Models: A Review," in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 11, no. 2, pp. 606-627, Feb. 2018, doi: 10.1109/JSTARS.2017.2781132.
[2] Shenglan Du, et al. "AdTree: Accurate, Detailed, and Automatic Modelling of Laser-Scanned Trees." *Remote Sensing* 2019, 11(18), 2074.
[3] Lowe, Thomas, and Joshua Pinskier. 2023. "Tree Reconstruction Using Topology Optimisation" *Remote Sensing* 15, no. 1: 172. https://doi.org/10.3390/rs15010172

2.15 Object segmentation from multiple-view point cloud renderings

The idea of this seminar is to use state-of-the-art general object segmentation approaches in 2D images such as Segment Anything Model and its follow-up improvements to segment desired objects from multiple individual viewpoints. We want to use the boundaries of these 2D segmentations and find cross sections of their projected outlines in 3D space, thus obtaining an approximate segmentation of the object in 3D space. We want to apply this to object segmentation in LiDAR point clouds.

References:

[1] Jain, A., Katara, P., Gkanatsios, N., Harley, A.W., Sarch, G., Aggarwal, K., Chaudhary, V. and Fragkiadaki, K., 2024. Odin: a single model for 2d and 3d segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 3564-3574).

[2] Wang, Y., Gong, M., Wang, T., Cohen-Or, D., Zhang, H. and Chen, B., 2013. Projective analysis for 3D shape segmentation. ACM Transactions on Graphics (TOG), 32(6), pp.1-12.

2.16 Interactive volume segmentation in VR

In this seminar, you will develop an interactive segmentation framework for volumetric datasets in virtual reality. The system will allow users to define segmentation regions by placing spatial control points directly within a VR environment. Each point is associated with a radius parameter that can be adjusted interactively, and a collection of such points defines a smooth implicit surface (such as metaballs or radial basis functions), that blends these control points into a continuous segmentation boundary. The resulting implicit surface defines the region of interest within the volume. You will implement a framework to render a volume and overlay user-defined segments, as well as real-time visual feedback as users add, remove, or modify control points. The system should support direct hand tracking for point placement and radius adjustment. You should assess the usability of the system for the task of generating detailed segmentation boundaries.

References:

[1] Blinn, James F. "A generalization of algebraic surface drawing." ACM transactions on graphics (TOG) 1.3 (1982): 235-256.

[2] Wyvill, Geoff, Craig McPheeters, and Brian Wyvill. "Data structure for soft objects." The visual computer 2.4 (1986): 227-234.

[3] Bloomenthal, Jules, and Ken Shoemake. "Convolution surfaces." Proceedings of the 18th annual conference on Computer graphics and interactive techniques. 1991.

2.17 Improvement of face-filling tools in Blender 3D software

When modeling 3D meshes, artists often remove individual vertices or face sets, creating gaps in the models. Blender already offers basic tools to fill these gaps with faces, such as the Fill and Grid Fill operators. However, these tools are ineffective on complex meshes with gaps that do not form perfect grids. The aim is to introduce new methods for filling gaps in 3D meshes. This topic is based on a Blender Summer of Code idea, but the scope should be narrowed to fit the seminar's length. The solution developed should ideally be submitted as a pull request to support Blender's open development, but it could also be expanded into a master’s thesis. The seminar requires a good understanding of Blender and the C++ programming language.

2.18 Replacement of 3D text representation in Blender 3D software

Blender is a 3D modelling software capable of manipulating various 3D representations. At present, 3D text is implemented through 3D curve evaluation, which is inefficient and does not integrate well with other tools, such as Geometry Nodes. The aim of the seminar is to replace the current implementation with a more efficient approach that supports rendering and editing of 3D text. The seminar is based on the Blender Summer of Code idea and should be adapted to fit the seminar's duration. Participants are expected to have a good understanding of Blender and the C++ programming language.